



Engineering Materials

Metals · Polymers · Ceramics · Composites

• MECHANICAL ENGINEERING • LESSON 02

Engineering Materials — The Foundation of Design

Every component a mechanical engineer designs must be made from something. The choice of material determines the component's **strength, weight, cost, manufacturability, and service life**. Get it wrong and the part fails; get it right and you build something that lasts decades. Materials selection is one of the most important and most underestimated skills in mechanical engineering.

The Four Classes of Engineering Materials



Figure 1 — The Four Classes of Engineering Materials

Key Mechanical Properties Every Engineer Must Know

Property	Symb ol	Definition	Unit	Why It Matters
Young's Modulus (Stiffness)	E	Slope of stress-strain curve in elastic region — resistance to elastic deformation	GPa	Deflection, vibration, resonance — how much a part bends under load
Yield Strength	σ_y	Stress at which permanent (plastic) deformation begins	MPa	Max safe working stress — designing to stay below this is essential
Ultimate Tensile Strength (UTS)	σ_u	Maximum stress before fracture	MPa	Safety factor calculations; minimum strength for structural parts
Elongation at Break (Ductility)	%	Percentage increase in length before fracture	%	Warnability before failure: >5% ductile, <2% brittle
Hardness	HV / HRC	Resistance to localised surface indentation	HV, HRC, HB	Wear resistance, machinability; hard parts resist scratch but may be brittle
Fatigue Strength	S_e	Stress level the material can endure for 10 ⁷ cycles without failure	MPa	Critical for rotating or cyclic-load parts: shafts, springs, connecting rods
Fracture Toughness	K _{Ic}	Resistance to crack propagation under load	MPa√m	Prevents catastrophic brittle fracture from small cracks or notches

Design rule: Always design so that the maximum working stress stays below yield strength ÷ safety factor. For static loads use SF = 1.5–2.0; for fatigue or dynamic loads use SF = 2.0–4.0.

The Stress-Strain Curve

The stress-strain curve is the single most important graph in materials engineering. It is obtained by pulling a standard test specimen in tension at a controlled rate and recording force vs extension. Every number in a material datasheet — yield strength, UTS, elongation, Young's modulus — is read directly from this curve.

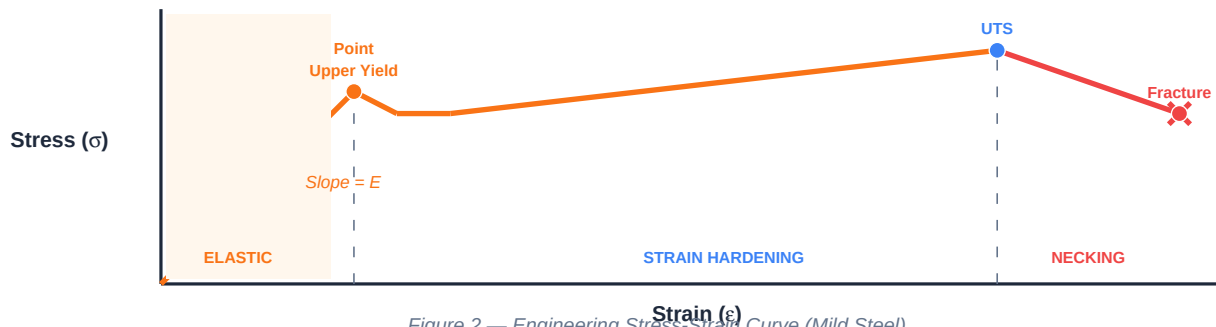


Figure 2 — Engineering Stress-Strain Curve (Mild Steel)

Common Engineering Metals and Their Properties

Material	Yield Str. (MPa)	UTS (MPa)	E (GPa)	Density (kg/m ³)	Key Use
Mild Steel (S275)	275	430	200	7 850	Structural frames, general fabrication
High-Strength Steel (S690)	690	770	200	7 850	Cranes, bridges, pressure vessels
Stainless Steel 316L	170	485	193	8 000	Chemical plant, food equipment, marine
Cast Iron (Grey)	~100	200	110	7 200	Engine blocks, machine bases (compressive)
Al Alloy 6061-T6	276	310	69	2 700	Aircraft, automotive, structural extrusions
Al Alloy 7075-T6	503	572	72	2 810	Aerospace, high-performance sports equipment
Ti-6Al-4V	880	950	114	4 430	Aerospace, medical implants, high-temp parts
Copper (annealed)	70	220	110	8 960	Electrical conductors, heat exchangers, plumbing

Heat Treatment of Steel

Process	Temperature	Cooling	Effect on Steel
Annealing	800–900°C	Furnace cool (very slow)	Softest, most ductile — relieves internal stress; easy to machine
Normalising	850–950°C	Air cool	Uniform grain; moderate strength; better than annealed for most uses
Hardening (Quenching)	800–900°C	Water or oil quench (fast)	Very hard, brittle martensite — must be tempered before use
Tempering	150–650°C	Air cool after hardening	Reduces brittleness of quenched steel; balance of strength + toughness

Case Hardening	900–950°C (carburising)	Quench outer case	Hard outer surface + tough core — gears, cams, crankshafts
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Martensite vs Pearlite: Fast quenching traps carbon in a strained lattice (martensite — hard and brittle). Slow cooling allows carbon to form pearlite (soft and tough). Tempering converts martensite partway toward a tougher microstructure without losing all the hardness.

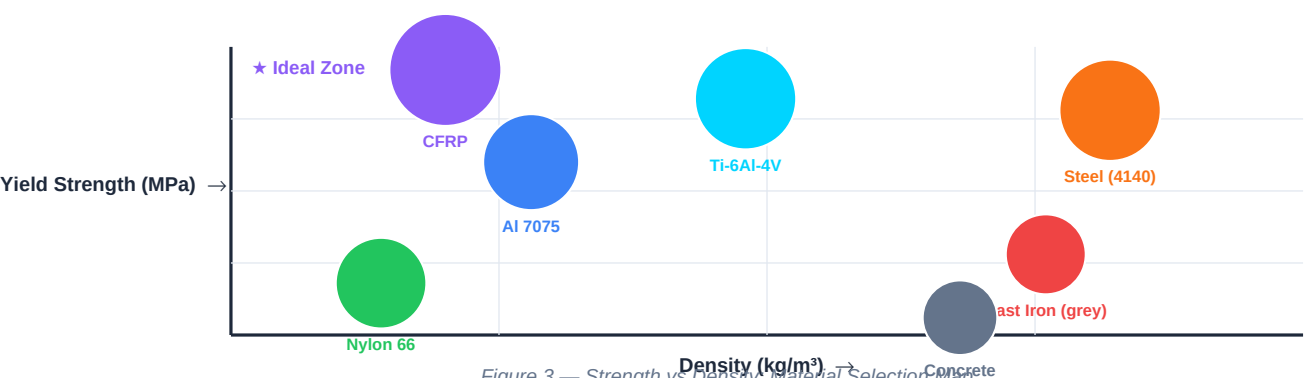
Polymers, Ceramics and Composites

Metals dominate traditional mechanical engineering, but modern design increasingly uses non-metals. **Polymers** save weight and resist corrosion. **Ceramics** operate at temperatures that would melt any metal. **Composites** combine materials to achieve properties impossible in any single material.

Class	Examples	Strengths	Weaknesses	Typical Application
Engineering Polymers	Nylon 66, PEEK, PTFE, Polycarbonate, ABS, HDPE	Low density, corrosion-free, self-lubricating, low cost	Low stiffness, creep under load, limited temp. range (<250°C)	Bearings, seals, gears, housings, pipes, lenses, 3D-printed parts
Ceramics & Glass	Alumina (Al ₂ O ₃), Silicon Carbide (SiC), Zirconia, Borosilicate glass	Extreme hardness, high temp. stability (>1 500°C), chemical resistance	Very brittle — zero ductility, low impact resistance, difficult to machine	Cutting tool inserts, turbine coatings, furnace linings, bearings at high temp.
Fibre Composites	CFRP (carbon fibre + epoxy), GFRP (glass fibre + epoxy), Kevlar composite	Highest strength-to-weight ratio; tailorable in any direction	Expensive to manufacture; difficult to repair; requires specialist joining	Aircraft structures, race car chassis, wind turbine blades, pressure vessels
Metal Matrix Composites (MMC)	Al-SiC, Ti-B4C, Cu-W	Metal toughness + ceramic stiffness; good at elevated temperature	High cost, complex processing	Brake discs, electronic packaging, aerospace structural components

Material Selection Map — Strength vs Density

The best material is not always the strongest — it is the one that delivers the required performance at minimum cost, weight, and risk. The chart below plots yield strength against density for common material families. The **top-left corner** (high strength, low density) is the "ideal zone" for weight-critical applications such as aerospace and automotive.



Design Requirement	Primary Selection Criterion	Top Material Candidates
Maximum stiffness per unit weight	Highest E/ρ (specific stiffness)	CFRP, Beryllium, Aluminium alloys
Maximum strength per unit weight	Highest σ _y /ρ (specific strength)	CFRP, Ti-6Al-4V, High-strength Al (7075)
High temperature operation (>600°C)	Melting point, creep resistance	Nickel superalloys, SiC, Al ₂ O ₃ ceramics
Corrosion resistance in seawater	Electrochemical potential + passivation	Ti-6Al-4V, 316L stainless, CFRP, HDPE

Wear-resistant sliding surface	Hardness + low friction coefficient	Case-hardened steel, PTFE, Nylon, Alumina
Low cost general structure	Yield strength per unit cost	Mild steel (S275/S355), standard Al 6061

What comes next — MECH-003: Engineering Mechanics — statics and dynamics, free body diagrams, equilibrium, moments, and the fundamental equations that let us calculate whether a structure or machine will hold together.